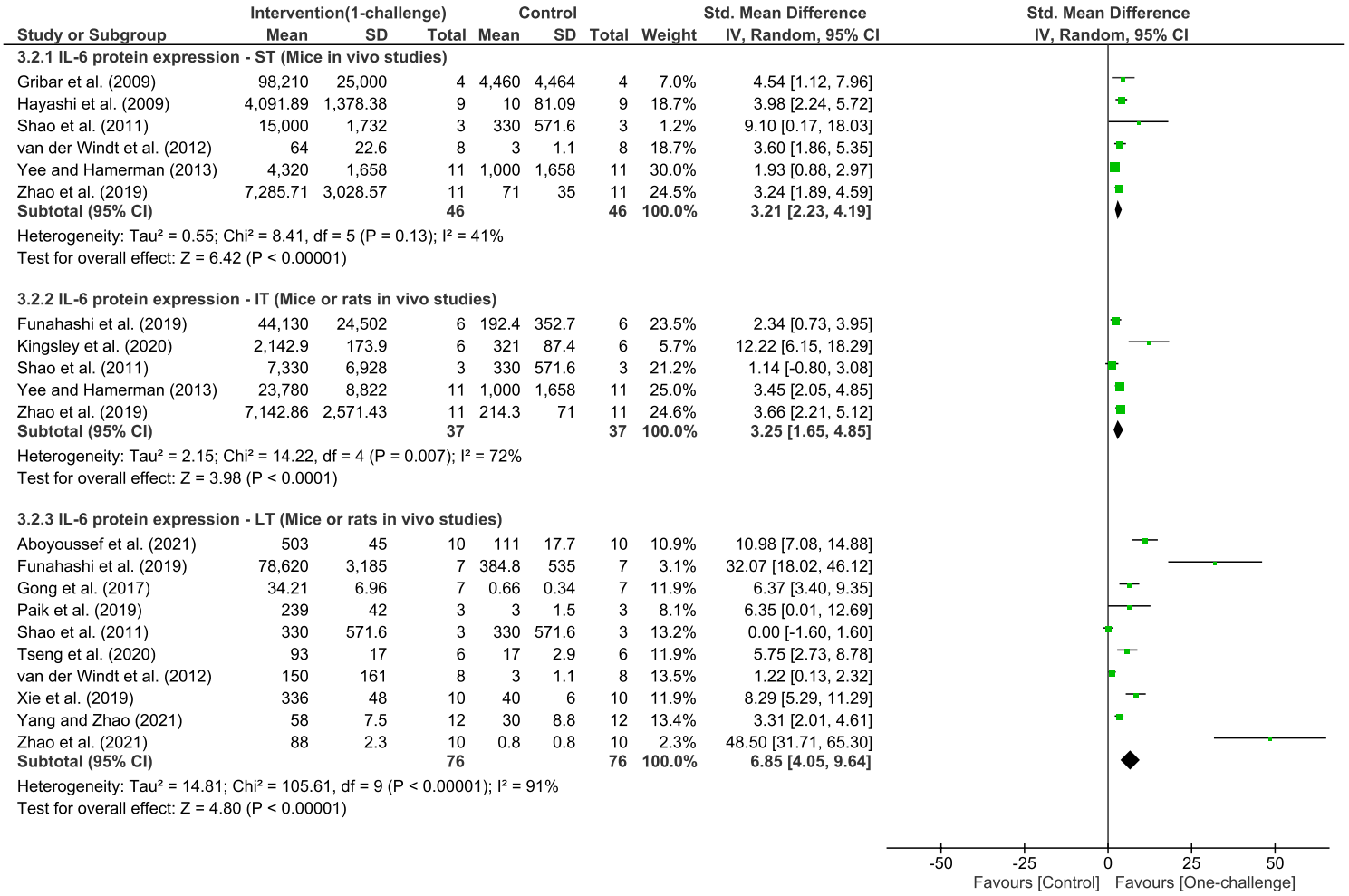

Intervention(1-challenge)
Control
Std. Mean Difference
Std. Mean Difference
Study or Subgroup
Mean SD Total Mean SD Total Weight IV, Random, 95% CI
IV, Random, 95% CI
3.2.1 IL-6 protein expression - ST (Mice in vivo studies)
Gribar et al. (2009) 98,210 25,000 4 4,460 4,464 4 7.0% 4.54 [1.12, 7.96]
Hayashi et al. (2009) 4,091.89 1,378.38 9 10 81.09 9 18.7% 3.98 [2.24, 5.72]
Shao et al. (2011) 15,000 1,732 3 330 571.6 3 1.2% 9.10 [0.17, 18.03]
van der Windt et al. (2012) 64 22.6 8 3 1.1 8 18.7% 3.60 [1.86, 5.35]
Yee and Hamerman (2013) 4,320 1,658 11 1,000 1,658 11 30.0% 1.93 [0.88, 2.97]
Zhao et al. (2019) 7,285.71 3,028.57 11 71 35 11 24.5% 3.24 [1.89, 4.59]
Subtotal (95% CI) 46 46 100.0% 3.21 [2.23, 4.19]
Heterogeneity: Tau² = 0.55; Chi² = 8.41, df = 5 (P = 0.13); I² = 41%
Test for overall effect: Z = 6.42 (P < 0.00001)
3.2.2 IL-6 protein expression - IT (Mice or rats in vivo studies)
Funahashi et al. (2019) 44,130 24,502 6 192.4 352.7 6 23.5% 2.34 [0.73, 3.95]
Kingsley et al. (2020) 2,142.9 173.9 6 321 87.4 6 5.7% 12.22 [6.15, 18.29]
Shao et al. (2011) 7,330 6,928 3 330 571.6 3 21.2% 1.14 [-0.80, 3.08]
Yee and Hamerman (2013) 23,780 8,822 11 1,000 1,658 11 25.0% 3.45 [2.05, 4.85]
Zhao et al. (2019) 7,142.86 2,571.43 11 214.3 71 11 24.6% 3.66 [2.21, 5.12]
Subtotal (95% CI) 37 37 100.0% 3.25 [1.65, 4.85]
Heterogeneity: Tau² = 2.15; Chi² = 14.22, df = 4 (P = 0.007); I² = 72%
Test for overall effect: Z = 3.98 (P < 0.0001)
3.2.3 IL-6 protein expression - LT (Mice or rats in vivo studies)
Aboyoussef et al. (2021) 503 45 10 111 17.7 10 10.9% 10.98 [7.08, 14.88]
Funahashi et al. (2019) 78,620 3,185 7 384.8 535 7 3.1% 32.07 [18.02, 46.12]
Gong et al. (2017) 34.21 6.96 7 0.66 0.34 7 11.9% 6.37 [3.40, 9.35]
Paik et al. (2019) 239 42 3 3 1.5 3 8.1% 6.35 [0.01, 12.69]
Shao et al. (2011) 330 571.6 3 330 571.6 3 13.2% 0.00 [-1.60, 1.60]
Tseng et al. (2020) 93 17 6 17 2.9 6 11.9% 5.75 [2.73, 8.78]
van der Windt et al. (2012) 150 161 8 3 1.1 8 13.5% 1.22 [0.13, 2.32]
Xie et al. (2019) 336 48 10 40 6 10 11.9% 8.29 [5.29, 11.29]
Yang and Zhao (2021) 58 7.5 12 30 8.8 12 13.4% 3.31 [2.01, 4.61]
Zhao et al. (2021) 88 2.3 10 0.8 0.8 10 2.3% 48.50 [31.71, 65.30]
Subtotal (95% CI) 76 76 100.0% 6.85 [4.05, 9.64]
Heterogeneity: Tau² = 14.81; Chi² = 105.61, df = 9 (P < 0.00001); I² = 91%
Test for overall effect: Z = 4.80 (P < 0.00001)
-50
-25
0
25
50
Favours [Control]
Favours [One-challenge]